

Defined Architecture of ANN Model

Project: Customer Churn Prediction

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1. Overview

The objective of this predictive modeling project is to forecast customer churn for a telecommunications company. An Artificial Neural Network (ANN) was selected as the modeling technique due to its ability to capture complex, non-linear relationships between customer attributes and churn behavior. The model is built using the Keras Sequential API (TensorFlow backend).

2. Network Architecture

The ANN architecture consists of one input layer, two hidden layers, and one output layer. The structure is designed to balance model complexity with generalization capabilities, preventing overfitting while ensuring high predictive accuracy.

2.1 Input Layer

- **Input Shape:** (10,)
- **Description:** The input layer accepts 10 standardized feature inputs derived from the dataset.
- **Features Included:** Demographic info (e.g., SeniorCitizen, Dependents), Services (e.g., PhoneService, Fiber optic), and Account details (e.g., Tenure, MonthlyCharges, Contract type).
- **Preprocessing:** All inputs are scaled using StandardScaler to ensure efficient gradient descent optimization.

2.2 Hidden Layers

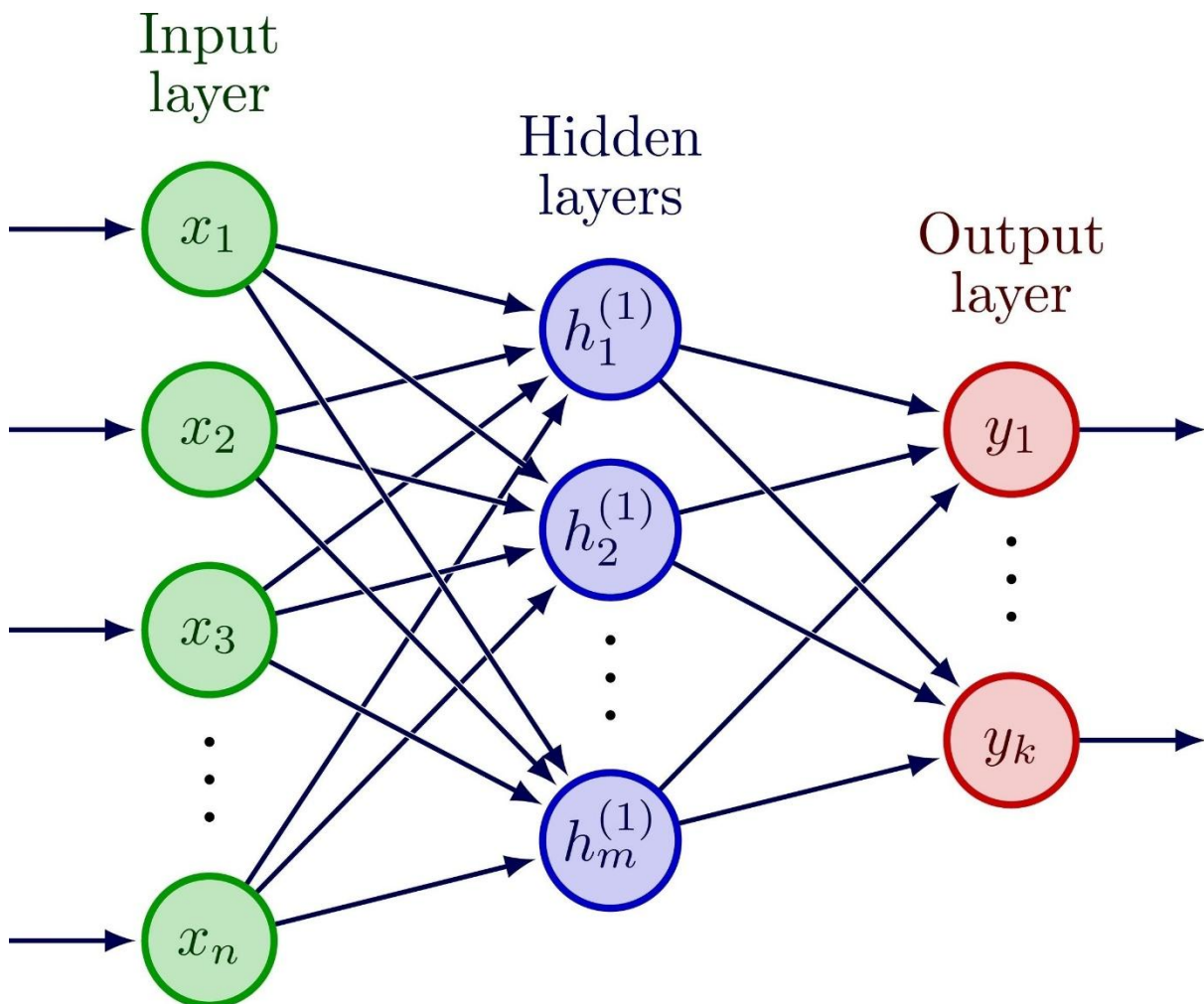
The model utilizes two hidden layers to extract features and learn patterns:

- **Hidden Layer 1:**
 - **Neurons:** 16
 - **Activation Function:** ReLU (Rectified Linear Unit)
 - **Reasoning:** The ReLU function is chosen to introduce non-linearity into the model, allowing it to learn complex boundaries between churners and non-churners. 16 neurons provide sufficient capacity to represent the feature space without excessive computational cost.

- **Hidden Layer 2:**
 - **Neurons:** 8
 - **Activation Function:** ReLU (Rectified Linear Unit)
 - **Reasoning:** This layer acts as a bottleneck, condensing the information learned from the first layer into higher-level abstract features before passing them to the output layer.

2.3 Output Layer

- **Neurons:** 1
- **Activation Function:** Sigmoid
- **Reasoning:** The Sigmoid activation function is essential for this binary classification task (Churn vs. No Churn). It squashes the output value between 0 and 1, representing the probability of a customer churning.
 - Output ≥ 0.5 : Predicted as Churn (1)
 - Output < 0.5 : Predicted as No Churn (0)



3. Model Optimization Strategy

To ensure robust performance and address specific data challenges, the following configuration was applied:

- **Optimizer:** Adam (Adaptive Moment Estimation). This optimizer was selected for its efficiency and ability to adapt the learning rate during training, leading to faster convergence compared to standard Stochastic Gradient Descent (SGD).
- **Loss Function:** Binary Crossentropy. This is the standard loss function for binary classification, penalizing predictions that deviate from the actual class labels.
- **Addressing Class Imbalance:**
 - **Issue:** The dataset contains significantly fewer churners than non-churners, which initially led to low Recall (the model struggled to identify actual churners).
 - **Solution: Class Weights** were computed and applied during training. This technique assigns a higher penalty to the model for misclassifying the minority class (Churners).
 - **Result:** This optimization significantly improved the model's sensitivity (Recall) for detecting potential churners, making the model more valuable for proactive retention strategies.